

ASSIGNMENT 2

Airport Ground Operations Management System

1. Scenario

You will design and build a system that manages ground-level airport operations for a single terminal: flight registration, gate assignment, passenger boarding, and baggage handling. The system should behave like a real tool used by ground staff and gate agents on a working shift — not a full airline reservation/booking platform.

Meridian Terminal Operations manages ground handling for a mid-sized airport terminal. Gate agents currently coordinate flight status, gate assignments, and boarding manually, using radios and paper boarding logs. Terminal management has hired your team to build the first version of a ground operations system — a console application that gate agents and ground staff can run to manage a shift's operations.

Many passengers at Meridian Terminal are connecting from an earlier flight rather than starting their journey there, popular flights are routinely oversold against a fixed number of seats, passengers frequently check in more than one bag, and ground staff are shared across multiple flights and gates during a single shift. The system must reflect these realities rather than treating every flight, passenger, and staff assignment as an isolated, one-off event.

Business Problem

Meridian Terminal currently has no reliable way to:

- Prevent assigning two flights to the same gate during overlapping time windows.
- Confirm a connecting passenger actually has enough time between their arriving flight and their next departure before letting them board.
- Track a passenger's total checked baggage weight across multiple bags on the same flight, rather than checking each bag in isolation.
- Accept bookings on a popular flight beyond its confirmed seat capacity, and fairly manage who boards if a seat later becomes available.
- Track how many hours a ground staff member has already worked across their assignments in a shift, to avoid over-assigning them.

System Goals

The system must allow ground staff to:

- Register flights and assign them to gates without conflicts.
- Register passengers, including an optional link to an earlier connecting flight, and verify sufficient connection time before boarding.
- Register baggage per passenger and track the combined weight of all their bags against the flight's allowance.
- Manage a flight's confirmed passenger list and a standby list once the flight is fully booked, promoting standby passengers as seats free up.
- Track each ground staff member's accumulated duty hours across their assignments and avoid over-committing them.

Actors / Users

- **Gate Agent** — checks passenger boarding eligibility, processes boarding.
- **Ground/Baggage Staff** — processes baggage, enforces weight restrictions.
- **Operations Supervisor** — registers flights, assigns gates and staff, updates flight status.
- **System** — enforces business rules and prevents invalid or unsafe operations.

Entities and Concepts Involved

The system should represent, at minimum, the following concepts (you decide the structure and relationships):

Entity	Notes
Flights	Distinguished by type (domestic/international), with a schedule, gate, seat capacity, and status. A flight may serve as the connecting (arrival) leg for a passenger continuing onward.
Gates	Defined availability window and restrictions (e.g., some gates may not support international flights).
Passengers	Distinguished by category (standard, VIP, reduced-mobility). A passenger may optionally be traveling on a connecting itinerary — arriving on one flight and departing on another.
Baggage	A passenger may check in more than one bag per flight; what matters for the allowance is the combined weight of all their bags, not any single bag alone.
Ground Staff	Assigned to flights/gates. Accumulates duty hours across all assignments during a shift.
Standby List	Passengers waiting for a seat on a flight that has reached its confirmed seat capacity, released in the order they were added as seats free up.

2. Constraints and Restrictions (required for consistent implementation)

- No persistence between runs (data lives only for the duration of the running session) unless you are separately instructed otherwise.
- **Design Decisions Left to You:** Some values are not fixed by this assignment on purpose — for example, the minimum connection time, the maximum staff duty hours per shift, the standby list capacity, and the baggage allowance per passenger category. You must choose a reasonable value for each and document your choice and reasoning in the README. There is no single correct number; what matters is that your choice is sensible and consistently applied.
- **Registering vs. Booking a Passenger:** Registering a passenger (adding their record to the system) and booking them onto a flight (confirming a seat or placing them on standby) are two separate operations. A passenger can be registered without yet having a confirmed booking. Design your classes and menu operations to reflect this separation.
- **Gate Occupancy Window:** The assignment refers to a flight occupying a gate during a time window, without defining exactly what that window is. You must decide how a flight's gate-occupancy window is determined (for example, from its arrival time to its scheduled departure, or some other consistent rule) and apply that definition consistently across your gate-conflict checks.

3. Requirements

The requirements are split into phases. Complete each phase, and ensure your implementation is testable against every requirement listed below.

Phase 1 — Functional Requirements

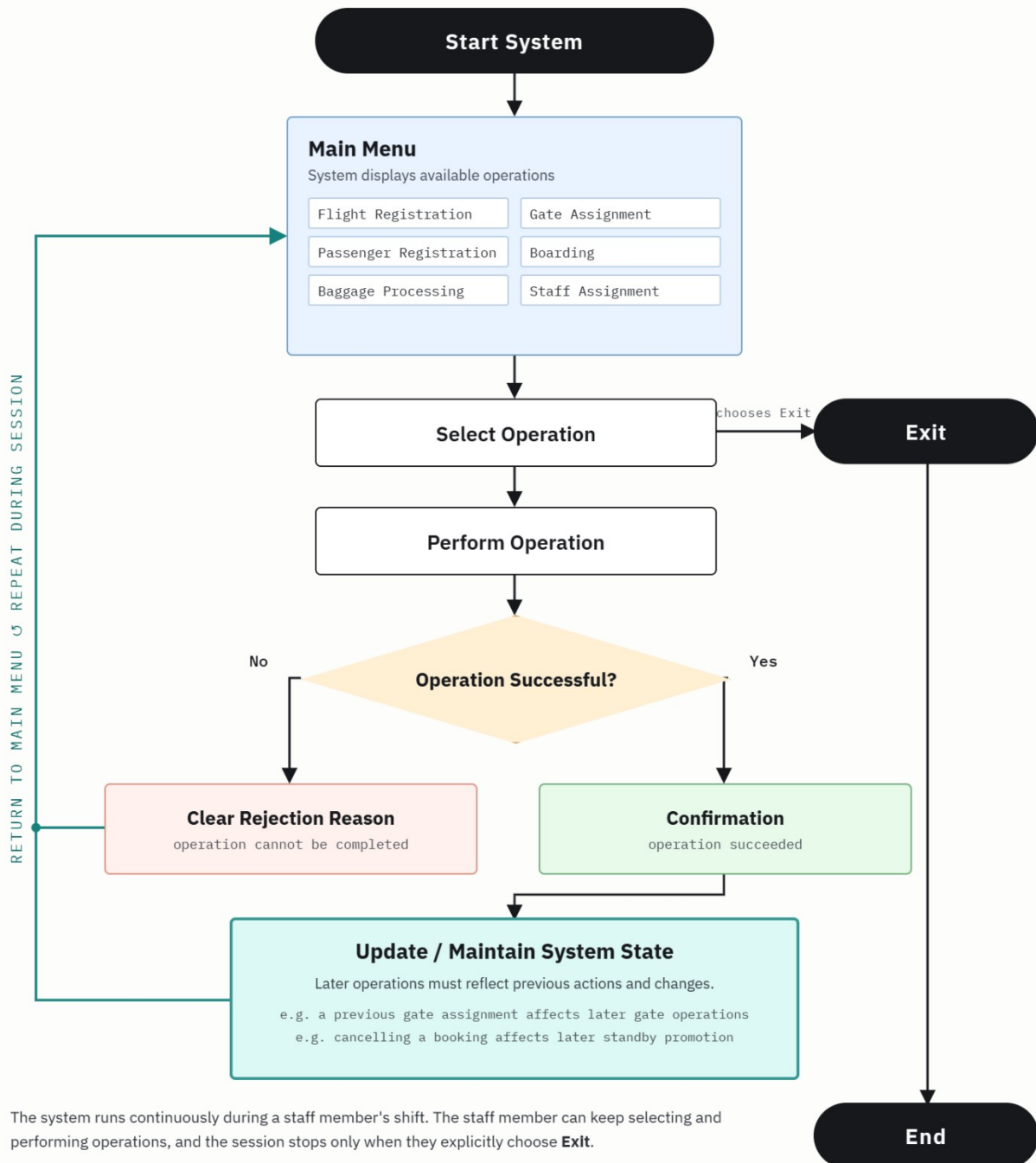
1. The system must allow registering a new flight with its type (domestic/international), scheduled time, gate requirements, and seat capacity.
2. **Gate Assignment Rule** - The system must allow assigning a flight to a gate. The system must prevent assigning a flight to a gate that is already occupied by another flight during an overlapping time window.
3. The system must allow registering a passenger against a flight. A passenger may optionally be linked to an earlier connecting flight arriving at the terminal.
4. **Connection Time Rule** - Before allowing a connecting passenger to board their next flight, the system must verify that enough time remains between the connecting flight's arrival and the next flight's departure, based on a minimum connection time you define. If there is not enough time, boarding must be denied with the specific reason.
5. **Cumulative Baggage Rule** - The system must allow registering baggage against a passenger, where a passenger may check in more than one bag for the same flight. The system must track the combined weight of all of that passenger's bags on that flight against the allowance for their category, and reject any bag that would push the running total over the allowance — even if that bag alone is within a normal single-bag limit.
6. **Standby Rule** - The system must allow booking a passenger onto a flight even after its seat capacity has been reached, placing them on a standby list instead of a confirmed booking.
7. **Standby Rule** - When a confirmed passenger's booking is cancelled, the system must automatically offer the freed seat to the earliest passenger on that flight's standby list.
8. **Staff Duty-Hour Rule** - The system must allow assigning a ground staff member to a flight or gate while tracking their cumulative duty hours across all assignments in the current shift. The system must reject a new assignment that would push their cumulative hours over a defined maximum, even if that single assignment looks short on its own.
9. **Flight Status Rule** - The system must allow updating a flight's status (e.g., delayed, boarding, departed, cancelled), and must prevent boarding or baggage operations inconsistent with that status.

AIRPORT GROUND OPERATIONS MANAGEMENT SYSTEM

Phase 2 – Main System Workflow

HIGH-LEVEL
OPERATIONAL WORKFLOW

 STAFF ACTION
 SYSTEM ACTION
 DECISION / RESULT
 SYSTEM STATE
 START / EXIT / END



Phase 3 — Main Menu & Detailed Operations

At minimum, the system must support the following operations:

Flight & Gate Management:

- Register a new flight
- Assign a flight to a gate
- Update flight status

Passenger & Boarding:

- Register a passenger (with an optional connecting-flight link)
- Check a connecting passenger's boarding eligibility
- Process boarding

Baggage:

- Register baggage for a passenger
- View a passenger's cumulative baggage weight for a flight

Booking & Standby:

- Book/confirm a passenger's seat on a flight
- Cancel a confirmed booking
- View a flight's standby list

Staff Management:

- Assign ground staff to a flight or gate
- View a staff member's cumulative duty hours

Phase 4 — Error Scenarios & Test Cases

The system must gracefully handle situations including, but not limited to:

- A staff member enters non-numeric input where a number is expected.
- An attempt to assign a flight to a gate that is already occupied during an overlapping window.
- An attempt to board a connecting passenger when the remaining connection time is below the minimum — should be rejected with the specific reason.
- An attempt to add a bag that would push a passenger's cumulative baggage weight over the allowance, even though that individual bag alone is within a normal limit — should be rejected.
- An attempt to book a passenger onto a flight whose standby list has also reached its own defined limit — should be rejected outright.
- Cancel a confirmed passenger's booking on an overbooked flight, and confirm that the earliest standby passenger is automatically promoted to a confirmed seat.
- An attempt to assign a staff member to a new task that would push their cumulative duty hours over the maximum, even though the new assignment alone looks short — should be rejected.
- An attempt to board a passenger or load baggage on a flight that has already departed or been cancelled.
- The system should give specific, meaningful feedback (not generic "error" messages) so staff understand exactly why an operation failed.

Phase 5 — Sample Input/Output Examples

```
=== Meridian Terminal – Ground Operations System ===
```

1. Register Flight
2. Assign Gate
3. Register Passenger
4. Check Boarding Eligibility
5. Register Baggage
6. Manage Bookings & Standby
7. Assign Staff
8. Exit

```
Select an option: 4
```

```
Enter Passenger ID: P-2210
```

```
Enter Next Flight Number: MT-315
```

```
Result: BOARDING DENIED
```

```
Reason: Only 25 minutes remain since the connecting flight's arrival; the minimum connection time is 45 minutes.
```

```
Select an option: 5
```

```
Enter Passenger ID: P-3391
```

```
Enter Flight Number: MT-210
```

```
Enter Baggage Weight (kg): 12
```

```
Result: BAGGAGE REJECTED
```

```
Reason: This bag would bring the passenger's total checked baggage to 35kg, exceeding the 30kg allowance for Standard passengers.
```

4. Deliverables

- Submit the complete Visual Studio / .NET solution/project source code, implemented in C# using OOP concepts, including classes and methods.
- Ensure the project builds and runs without errors.
- Include a short README describing how to run the program and a summary of implemented features.
- **Submission Instructions:** Submit a ZIP file containing your complete project. The ZIP file name must be your email address.